NHH

HOME EXAMINATION

BAN404

Spring, 2021

Date: 10.06.2021

Time: 9.00-17.00

Number of hours: 8

THE HOME EXAMINATION SHOULD BE SUBMITTED IN WISEFLOW

You can find information on how to submit your paper here:
https://www.nhh.no/en/for-students/examinations/home-exams-and-assignments/

Your candidate number will be announced on StudentWeb. The candidate number should be noted on all pages (not your name or student number). In case of group examinations, the candidate numbers of all group members should be noted.

Collaboration between individuals or groups on submission preparation, as well as exchange of self-produced materials between individuals or groups is prohibited. The answer paper must consist of individual's or the group's own assessments and analysis. All communication during the home exam is considered cheating. All submitted assignments are processed in Urkund, a plagiarism control system used by NHH.

SUPPLEMENTARY REGULATIONS TO HOME EXAMINATIONS

Find more information under chapter 4.4 in the regulations for fulltime study programmes

Regulations for Full-time Study Programmes at the Norwegian School of Economics (NHH) | NHH

All pages, including front page: 4

Instructions

Your solution should be submitted as two files.

1. A pdf-file where both your R-code and the answers to the tasks are given.
2. Either a well documented R-script (.R) or an R markdown file (.Rmd) which can reproduce the results in the pdf-file.

The datasets used here are well analyzed before. Make sure to refer to other's work if you use it to solve the tasks.

A general recommendation: Be pragmatic if you run into difficulties. You need to find models that can produce predictions. Don´t let the perfect prediction stand in the way for a good one!

The maximum total score on the exam is 100 and the number in brackets after each subtask shows the maximum score for that subtask.

Task 1

In this task you will look at the `Smarket` data in the `ISLR` package.

a) Use the bootstrap to create a histogram of the sampling distribution for the volatility of the S&P stock index, i.e. the standard deviation of the returns. Observations of the returns are in the variable `Today`. Start by setting the seed to 1, `set.seed(1)`. Use 1000 bootstrap replicates. (10)
b) Compute a 95% confidence interval based on the bootstrap, assuming that the volatility is normally distributed. (4)
c) Compute a 95% confidence interval based on the bootstrap, not making the normality assumption but use percentiles from the bootstraped volatilites. (5)
d) Squared returns, `Today^2`, can be used as a proxy for volatility. Squared returns are often autocorrelated, indicating that volatility can be predicted. Run a regression of the square of `Today` on the square of `Lag1` and interpret the estimate of the regression parameter for the square of `Lag1`. (5)
e) Compute the bootstraped standard error of the regression coefficient in front of the square of `Lag1` in the model in d) and compare it with the standard error from the regression output. (Hint: An estimated regression coefficient from a simple linear regression with intercept made by `reg=lm(y~x)` can be obtained by `reg$coefficients[2]`). (10)

Task 2

In this task you will look at the data set `dataCar`, vehicle insurance policies, in the `insuranceData` package. You will find a description of the data set using the help function `help(dataCar)` and the link to the source for the data mentioned there. `exposure` is a measure of the risk the insurer takes on for a insurance taker. You should predict the claim amount, `claimcst0`, given that an insurance policy has at least one claim.

a) Start by first selecting the observations where there is an occurrence of claim, `clm!=0`. Also, remove the variable `X_OBSTAT_`. Work with this new data set for the rest of Task 2. (3)

b) Use descriptive statistics, the description of the data in the help-function and common sense to remove variables that you think will not be helpful. Motivate your choices thoroughly. Also, make sure that all variables are on the right format. (7)

c) Use a cross validation approach to evaluate predictions for the following models: A linear regression with all predictors left in the dataset after 2b, all possible simple (one-predictor) linear regression models and finally, a model without any predictors (a linear regression of `claimcst0` on an intercept). You should use a CV method that estimates the `testMSE` as precisely as possible at the same time as being feasible to implement. Motivate your choice of CV-method. If random draws are, somehow, used in your analysis, first set the seed to 123, `set.seed(123)`. (14)

d) Use a generalized additive model (GAM) with smoothing splines for the numerical variables. Include categorical variables in the model as well. You can use the `s`-function in the `gam`-library, with argument `df=4`, without motivation. Plot the result and comment on the relationship between the predictors and the dependent variable. (7)

e) Use your conclusions in 2d to specify an appropriate GAM-model. Compute the `testMSE` for this GAM-model. (4)

Task 3

In this task we will continue to work with the `dataCar` data but now focus on predicting the variable `clm`. Note that the starting point is the original `dataCar`, not the version you created in Task 2.

a) Start by removing the variables `X_OBSTAT_`, `numclaims` and `claimcst0`. Compute and interpret appropriate cross-tables of `clm` and the categorical variables. Also compute and interpret well-chosen descriptive statistics (such as central tendencies and measures of variation) for the numerical variables for each of the two categories of `clm`. Are some of the categories of the categorical variables more interesting than others? Based on this, do you see any promising predictors for `clm`? Mention a drawback with such pairwise comparisons. (7)

b) Based on your hypotheses from 3a, use logistic regression based on all or a subset of the predictors to predict `clm`. Interpret the estimated coefficients. (7)

c) Modify the model if you believe it to be sensible. Use the validation set approach with a 50/50 training/test split to evaluate the predictions from the logistic regression. Choose a threshold for when an estimated probability should lead to a prediction of a claim. You do not have to use a formal method but imagine a decision problem, using the prediction, that relies on the assumption that it costs more to predict a claim as a "non-claim" than vice versa. Compute a confusion matrix and argue for your choice of threshold in the context of a trade-off between false positive and false negative predictions. Set the seed to 1234, `set.seed(1234)`. (10)

d) Use boosted trees to predict `clm` and evaluate the predictions with the same approach as in 3c. (10)